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April 2022

**B.Tech. (CE-DS/CE/CSE/IT/CSE-ML) - I SEMESTER
Mathematics-I (Calculus & Linear Algebra) (BSC-103E)**

Time: 3 Hours] [Max. Marks: 75

Instructions:

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART -A

1. (a) Find the area under the curve $y = \frac{\log_e x}{x^2}$ from $x = 1$ to $x = \infty$. (1.5)
- (b) Evaluate $\int_0^1 x^{11} (1-x)^5 dx$. (1.5)
- (c) Evaluate $\lim_{x \rightarrow \infty} \frac{x^n}{e^x}$ if it exists, where n is a positive integer. (1.5)

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(d) Find c of Lagrange's mean value theorem for

$$f(x) = x^3 + x \text{ in } [1, 2]. \quad (1.5)$$

(e) If $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$, then show that AA^* is

Hermitian. (Here A^* denotes the transpose of the conjugate of A). (1.5)

(f) If ω is a complex cube root of unity then evaluate

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^4 \end{vmatrix}. \quad (1.5)$$

(g) Check whether the vectors $(1, 0, -1)$, $(2, 5, 1)$, $(0, -4, 3)$ in $\mathbb{R}^3$ are linearly independent or not? (1.5)

(h) The product of two eigen values of the matrix

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

is 16. Find the third eigen value. (1.5)

(i) Define orthogonal matrix and also give one example. (1.5)

(j) If $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is a linear transformation such that Nullity of $T = 1$. Find Rank of T . (1.5)

PART -B

2. (a) Find equation of evolute of the cycloid
 $x = a(\theta - \sin \theta), y = a(1 - \cos \theta).$ (8)

- (b) Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, where $\Gamma(n)$ denotes gamma function with parameter n . (7)

3. (a) The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about x -axis to generate a solid. Find the volume of the solid. (8)

- (b) Using Taylor's theorem, prove that

$$x - \frac{x^3}{6} < \sin x < x - \frac{x^3}{6} + \frac{x^5}{120} \text{ for } x > 0. \quad (7)$$

4. (a) A cone circumscribed a sphere of radius r . Prove that when the volume of the cone is minimum, its height is $4r$ and the semi-vertical angle is $\sin^{-1}\left(\frac{1}{3}\right)$. (10)

- (b) Find the value of k so that the rank of the matrix

$$\begin{bmatrix} 6 & 3 & 5 & 9 \\ 5 & 2 & 3 & 6 \\ 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & k \end{bmatrix} \text{ is 3.} \quad (5)$$

5. (a) Reduce the set of equations $x^2yz = xy^2z^3 = x^3y^2z = e$ into a system of linear equations and hence, solve it. (9)

- (b) Find the linear transformation $T : \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$

determined by the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ in the standard basis. Also, evaluate $T(-2, 2, 3)$ and $T(1, 0, -2)$. (6)

6. (a) For the linear transformation $T : \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y) = (x + y, x - y, y)$, find the basis and dimensions of

(i) its range space;

(ii) its null space. Also verify Rank-Nullity Theorem. (9)

- (b) Verify Cayley Hamilton theorem for the matrix

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

and hence find A^{-1} . (6)

7. Find an orthonormal basis of the inner product space $\mathbb{R}^3(\mathbb{R})$ with the standard inner product, given the basis $\{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$ using Gram-Schmidt process. Also, find the coefficients of the vector $(1, 1, 2)$ relative to the orthonormal basis. (15)

